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Semester: 4

DESIGN A LOW-COST SCISSOR LIFT FOR INDUSTRIAL APPLICATIONS

By

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1 Introduction

Scissor lifts serve as indispensable tools for various industrial tasks. These mobile platforms, characterized by their distinct scissor-like structure, provide a secure and effective means of elevating workers and equipment to elevated locations. Whether it's installing ceilings or repairing lights, they prove to be versatile assets capable of accommodating a wide array of tasks.

However, amidst the perpetual concern over costs, there exists a continuous drive to enhance these machines. This report delves deeply into the realm of scissor lifts, examining their utilization, requisite safety protocols, and potential for cost-effective design improvements. By amalgamating insights from experts and real-world scenarios, the objective is to identify strategies for reducing the production expenses of scissor lifts while upholding their operational efficiency.



The overarching goal is to develop an innovative and economically viable scissor lift tailored to the diverse demands of industrial environments. This endeavor promises heightened productivity, reinforced safety standards, and enhanced efficiency in working at elevated heights. In essence, it revolves around the enhancement of industrial practices for optimal outcomes.

2 Market Survey

There are five main types of scissor lifts available in the market, each suited for specific needs:

1. Hydraulic Scissor Lifts



- These are the most common type of scissor lift and utilize a hydraulic system to raise and lower the platform.
- They are known for their stability and lifting capacity and are ideal for indoor and outdoor use on flat, even surfaces.

Capacity: 200 - 2000kg

Lifting height: 4 -18m

Price: LKR 196,119.20 - LKR 980,596.00

Driving mode: hydraulic drive

Power supply:

- Diesel Engine
- Diesel Engine
- Three phase current

Travel methods:

- Manual Push/Pull (for smaller lifts)
- Self-propelled with wheels (electric or hydraulic drive)
- Vehicle Mounted (lifts permanently mounted on a truck bed)

	<u>Special features:</u> • Detachable Guardrails (typically 1 meter height) • Easy to Transport and Use • Durable Construction (steel or manganese steel main structure) • Safe Operation: ○ 24V control voltage ○ Gravity lowering (platform descends using its own weight) ○ Optional Features: * Battery drive for increased outdoor flexibility * Solid tires for better durability (compared to pneumatic tires) <u>Design shortcomings:</u> • Relatively High Cost Compared to Other Types (like pneumatic scissor lifts) • May Require Additional Power Unit for Increased Capacity (in rare cases) • Lifting Speed Variations (might differ depending on load weight) • <u>Solution/Benefit:</u> <u>Solution idea:</u> • Safe lowering even in power failure due to gravity-based descent mechanism
2. Electric Scissor Lifts 	• As the name suggests, electric scissor lifts are powered by batteries, making them ideal for indoor applications where noise and fume control are crucial. • They are typically smaller and lighter than hydraulic models, with lower weight capacities. <u>Capacity:</u> 1200kg <u>Lifting height:</u> 4 -12m

	<u>Price:</u> LKR 705 970 – LKR 1 227 774 <u>Driving mode:</u> Electrical hydraulic motor drive <u>Power supply:</u> • AC Power (for lifting operation) • DC Battery Power (for self-propulsion) <u>Travel methods:</u> • Manual Push/Pull (for smaller lifts or positioning) • Self-propelled with wheels (powered by DC battery) <u>Special features:</u> • Extended Battery Life (compared to hydraulic lifts) • Anti-Slip Gear System (prevents unintended platform movement) • Platform Extension (increases work area) • Off-Slab Mobility (suitable for some uneven surfaces) • Single Operator Control (controls located on platform for independent operation) • Zero Emissions (ideal for indoor use) • Low Noise Level (promotes a safer and quieter work environment) • Suitable for Indoor and Specific Outdoor Applications (where fumes and noise are restricted) • All-Wheel Drive (on some outdoor models for better maneuverability) <u>Design shortcomings:</u> • Limited to Flat Surfaces (not suitable for significant inclines or uneven terrain) • Battery Runtime Limitations (may require recharge breaks during extended use)
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	<u>Solution idea:</u> Extended battery life compared to hydraulic lifts reduces downtime for recharging.
3. Diesel Scissor Lifts 	- Diesel scissor lifts are powered by diesel engines, making them suitable for outdoor construction sites and uneven terrain. - They offer superior lifting capacities and can reach higher working heights compared to electric models. <u>Capacity:</u> 1500kg - 3500kg <u>Lifting height:</u> 6 -18 m <u>Price:</u> LKR 1 074 302 – LKR 3 069 437 <u>Driving mode:</u> - Hydraulic Drive (powered by diesel engine) <u>Power supply:</u> - Diesel Engine (provides power for lifting and driving) <u>Travel methods:</u> - Self-propelled with wheels (powered by the diesel engine) <u>Special features:</u> - Superior Lifting Capacity (compared to electric models) - Excellent for Outdoor Use: - Handles uneven terrain and rough surfaces - More powerful engine for demanding applications

	<u>Design shortcomings:</u> • Higher Emissions (not suitable for enclosed spaces) • Louder Noise Levels (compared to electric models) • Bulkier Size (may require more space for storage and transportation) <u>Solution idea:</u> • Ideal for construction sites, outdoor maintenance, and other applications requiring heavy lifting and off-road capabilities.
4. Rough Terrain Scissor Lifts 	• These robust scissor lifts are specifically designed for uneven and off-road conditions. • They have large tires with high ground clearance and four-wheel drive for added stability and traction on rough surfaces. <u>Capacity:</u> 3,700kg (common capacity, may vary depending on model) <u>Lifting height:</u> 10 -18 m <u>Price:</u> LKR 3 069 437 – LKR 12 277 748 <u>Driving mode:</u> • Hydraulic Drive (powered by diesel engine) <u>Power supply:</u> • Diesel Engine (provides power for lifting and driving)

	<u>Travel methods:</u> • Manual Push/Pull (for limited maneuvering) • Self-propelled with wheels (powered by the diesel engine) <u>Special features:</u> • High Load Capacity (handles heavier loads compared to other scissor lifts) • Excellent Off-Road Performance: ○ Superior traction control for uneven terrain ○ High ground clearance for obstacles • Significant Lifting Height (suitable for reaching elevated work areas outdoors) • Expandable Outriggers (improve stability on uneven surfaces) • Large Platform Workspace (increases work area) • Dual Controls (ground and platform control for operator convenience) • Platform Extension (further expands work area) • Enhanced Safety Features: ○ Guardrails for operator protection
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	○ Emergency lowering system ○ Double-acting lift cylinders with lock valves <u>Design shortcomings:</u> • Not Suitable for Indoor Use (due to diesel engine emissions) • High Emissions (contribute to air pollution) <u>Solution idea:</u> Ideal for construction sites, landscaping projects, and other outdoor applications requiring high lifting capacities and off-road capabilities.
5. Pneumatic Scissor Lifts 	• Pneumatic scissor lifts utilize compressed air for operation. • They are lightweight and compact, making them suitable for confined spaces or delicate flooring. • However, they typically have lower lifting capacities and platform heights compared to other types. <u>Capacity:</u> Typically lower than other scissor lifts, around 200lbs (may vary depending on model) <u>Lifting height:</u> Typically shorter than other scissor lifts, with a common range of up to 29 inches (may vary depending on model) <u>Price:</u> LKR 276 249 – LKR 306 943 <u>Driving mode:</u>

	• Powered by compressed air (shop air), eliminating the need for electricity or hydraulic fluid. <u>Power supply:</u> • Powered by compressed air (shop air), eliminating the need for electricity or hydraulic fluid. <u>Travel methods:</u> • Stationary (for lightweight, compact models) • Mobile (with wheels for manual pushing/pulling) <u>Special features:</u> • Cleanliness: Ideal for sensitive environments like clean rooms and food processing plants due to the absence of oil and electrical components. • Safety: Minimizes risks associated with hydraulic leaks or fluid contamination. • Portability: Some models are compact and lightweight, making them easier to maneuver within a workspace. • Simple Maintenance: Air-powered design generally requires less maintenance compared to hydraulic systems. <u>Design shortcomings:</u>
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	• Limited Lifting Capacity: Compared to hydraulic lifts, pneumatic models generally have a lower maximum weight capacity. This makes them unsuitable for heavy-duty lifting applications. • Noise: The operation of compressed air cylinders can be quite noisy. In some environments, this might necessitate soundproofing measures or using the lift in designated areas. • Controllability: Air's compressibility can make achieving very precise height adjustments slightly more challenging compared to hydraulic systems. However, with practice, operators can become adept at controlling the lift. • Air Consumption: Depending on the frequency of use and lifting height, pneumatic scissor lifts can consume a significant amount of compressed air. This might necessitate a powerful compressor system or impact other air-powered tools in the workspace. • Susceptibility to Moisture: Moisture buildup in the compressed air system can lead to corrosion and potential malfunctions. Proper air filtration and drainage are crucial for maintaining the lift's performance and longevity. <u>Solution idea:</u> 1. Lifting Capacity:
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	Hybrid Designs: Develop hybrid lifts that combine compressed air with a smaller hydraulic component for situations requiring a slight increase in lifting capacity without sacrificing the environmental benefits of pneumatics. 2. Noise: Quieter Compressors: Invest in research and development of quieter compressed air technology specifically for powering scissor lifts. Enclosed Designs: Explore incorporating soundproofing materials into the scissor lift structure to dampen noise generated by the cylinders. 3. Controllability: Advanced Control Systems: Develop computer-aided control systems that compensate for air compressibility, offering more precise height adjustments. 4. Air Consumption: Energy-efficient Designs: Optimize the design of the pneumatic system to minimize air consumption while maintaining operational efficiency. Variable Speed Control: Implement variable speed control features for the lift, allowing for slower, more precise movements that consume less air. 5. Moisture Susceptibility: Improved Filters and Drainage: Develop and implement more efficient air filtration systems and
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	drainage mechanisms to remove moisture before it enters the lift's components. Corrosion-resistant Materials: Use corrosion-resistant materials in critical parts of the lift to minimize damage from moisture buildup.
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Scissor Lift Type	Pros	Cons
Hydraulic	Stable and high lifting capacity	Slower operation in colder weather
Diesel	Superior lifting capacity, can reach a height of 30 feet, with advanced models extending as high as 60 feet	Noisy, emits fumes, best suited for outdoor use
Electric	Quiet, fume-free, less bulky than diesel lifts	Limited by battery life
Rough Terrain	High load capacity	Not suitable for indoor use
Pneumatic	Clean (no oil or electrical components)	Requires air pressure, may not be as powerful as other types

3 Main design features and the major customer requirements

Main Design Features:

- **Lifting Mechanism:** This is the core functionality, typically achieved through hydraulics (using oil), electric motors, or compressed air.
- **Platform:** The work area for personnel and materials. Size, weight capacity, and extension capabilities are important factors.
- **Base and Chassis:** Provides stability and houses the power source and drive system. May include wheels for self-propulsion or outriggers for additional stability.
- **Control System:** Enables safe and efficient operation, including raising/lowering, steering (if self-propelled), and emergency stop functions.
- **Safety Features:** Guardrails, tilt sensors, overload protection, and emergency lowering mechanisms are crucial.

Major Customer Requirements:

- **Cost:** A key factor, especially for industrial applications where budget constraints exist.
- **Lifting Capacity:** The weight the scissor lift can safely handle.
- **Lifting Height:** The maximum working height required for specific tasks.
- **Portability and Maneuverability:** Ease of movement within the workspace, especially for smaller models.
- **Durability and Reliability:** Ability to withstand frequent use and harsh industrial environments.
- **Safety:** Compliance with safety regulations and features that minimize operator risks.
- **Ease of Use:** Intuitive controls and simple operation for efficient work.
- **Environmental Impact:** Noise levels, emissions (if applicable), and energy efficiency are becoming more important considerations.

Shortcomings of Existing Designs (based on market survey findings):

- **Hydraulic Lifts:**
 - Relatively high cost compared to other types.

- May require additional power units in rare cases.
- Lifting speed variations depending on load weight.
- Electric Lifts:
 - Limited to flat surfaces, not suitable for uneven terrain.
 - Battery runtime limitations may require recharge breaks during extended use.
- Diesel Lifts:
 - Higher emissions restrict use in enclosed spaces.
 - Louder noise levels can be a safety concern and disrupt work environments.
 - Bulkier size requires more storage and transportation space.
- Rough Terrain Lifts:
 - Not suitable for indoor use due to diesel engine emissions.
 - High emissions contribute to air pollution.
- Pneumatic Lifts:
 - Limited lifting capacity restricts heavy-duty applications.
 - Noisy operation might necessitate soundproofing measures.
 - Air compressibility can make precise height adjustments more challenging.
 - High air consumption requires a powerful compressor system.
 - Susceptibility to moisture buildup can lead to corrosion and malfunctions.

4 Proposed Designs

No	Conceptual Design	Description
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1		Hybrid Lifting System Scissor Lift: ✓ This design combines electric motor with a small two hydraulic cylinder. ✓ Electric motor provide quiet operation. ✓ The small hydraulic cylinders offers a boost in lifting capacity compared to purely electric models, catering to a wider range of industrial needs without the high cost of a full hydraulic system. ✓ The design incorporates longer beams that extend as the scissor lift elevates. These longer beams enhance stability, especially when working at maximum platform heights. ✓ The platform can be designed with an interchangeable safety bench, providing additional protection for workers while performing tasks at elevated positions. This safety bench can be easily attached or detached depending on the specific application and risk assessment.
2		Improved Pneumatic Scissor Lift: ✓ This design focuses on improvements to existing pneumatic scissor lifts. ✓ It implements a two-stage air filtration system to remove moisture more effectively, reducing corrosion risks and extending lift life. ✓ An automatic condensate drain is integrated to prevent moisture buildup within the system. ✓ The cylinder design is optimized, potentially using lightweight aluminium cylinders, or exploring alternative configurations to improve lifting capacity without significantly increasing complexity.
		Simplified Electric Scissor Lift with Manual Propulsion: ✓ This design prioritizes a low-cost and portable solution. ✓ It utilizes a compact and efficient electric motor directly powering the scissor mechanism for lifting. ✓ The self-propulsion system is eliminated and replaced with a robust handle system for manual maneuvering. ✓ The chassis and platform are constructed using lightweight materials like aluminum tubing. ✓ This design caters well to indoor environments with smaller work areas where maneuverability is less critical.

		Height-Adjustable Manual Scissor Lift with Counterweight: ✓ This proposal offers the most basic and cost-effective scissor lift design. ✓ It implements a hand crank or screw jack mechanism to raise and lower the platform. ✓ The scissor mechanism has multiple locking positions to achieve different platform heights. ✓ A counterweight system on the base improves stability during operation, especially at higher platform heights. ✓ This design caters to situations where low cost and portability are paramount, such as light maintenance tasks or reaching moderately high work areas.
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5 Best Design

Criteria	Weighting (Out of 10)	Hybrid Lifting System	Improved Pneumatic	Simplified Electric (Manual Propulsion)	Height- Adjustable Manual
Cost-effectiveness	8	7	6	9	10
Functionality	7	8	5	3	2
Safety	10	9	8	7	6
Ease of Use & Maintenance	7	8	9	10	10
Environmental Impact	5	4	5	5	5
Total weight	37	36	33	34	33

Weighting Explanation:

- **Cost-effectiveness (High Weight):** This is likely a major concern for many industrial buyers.
- **Functionality (High Weight):** The design should meet a range of lifting needs.
- **Safety (Highest Weight):** This is paramount for any scissor lift.
- **Ease of Use & Maintenance (Medium Weight):** Balancing user-friendliness with maintenance requirements.
- **Environmental Impact (Medium Weight):** While not the top priority, quieter operation and lower emissions are becoming more important.

Evaluation:

- **Hybrid Lifting System:** Scores well in most categories but might have a slightly higher cost due to the combined system.
- **Improved Pneumatic:** Offers better lifespan and potentially higher capacity but adds complexity.
- **Simplified Electric (Manual Propulsion):** Most cost-effective and easy to use, but limited to smaller areas and tasks.
- **Height-Adjustable Manual:** Lowest cost and simplest design, but limited functionality and requires manual effort.

Interpretation:

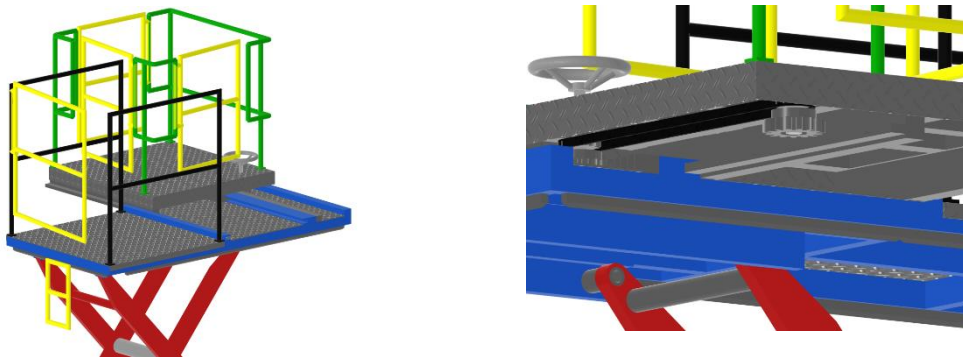
- The **Simplified Electric (Manual Propulsion)** design seems most aligned with the focus on low cost. Therefore this might be a good choice.
- The **Hybrid Lifting System** offers a good balance of features and cost, but further modification might be needed.

6 Best Final Design

I select the final best design as an enhanced version of Proposalv1: Hybrid Lifting System Scissor Lift. This new design incorporates additional features and functionalities, making it a versatile and user-friendly solution for industrial applications.

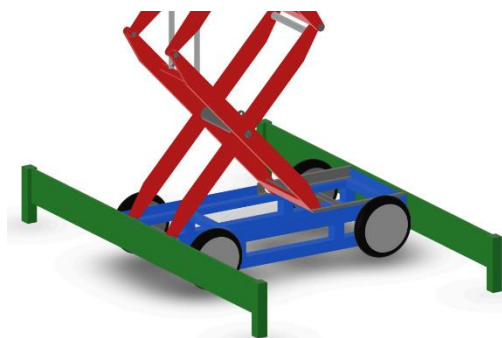
Enhanced Features:

- **Horizontal Platform Movement:**



The top platform integrates a hand crank mechanism that allows for precise horizontal movement. This eliminates the need to reposition the entire scissor lift for minor adjustments, increasing efficiency and user convenience. The hand crank connects directly to the platform, providing a simple and reliable method for horizontal control.

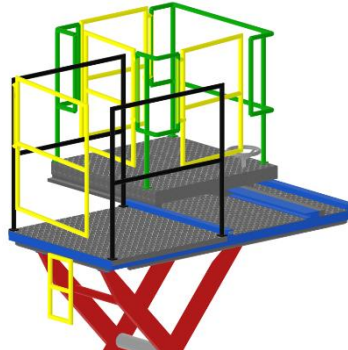
- **Improved Stability:**



To ensure stability, especially at maximum lifting heights, long outriggers are attached to the scissor lift base. These outriggers extend outwards as the platform elevates, significantly increasing the base footprint. The outrigger design prioritizes stability while remaining

detachable. By loosening designated screws, the outriggers can be easily removed for compact storage and transportation.

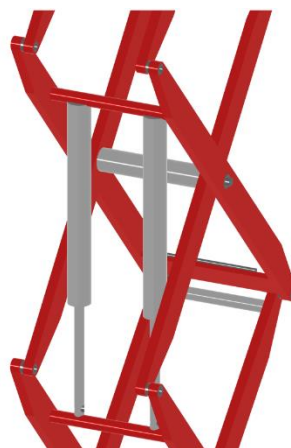
- **Fixed Safety Bench:**



A fixed safety bench with integrated guardrails is a permanent fixture on the platform. This design prioritizes worker safety by providing continuous fall protection throughout operation. The fixed bench eliminates the need for additional setup or removal steps, simplifying the overall process.

- **Hybrid Power System:**

The core of this design is a hybrid power system that combines an electric motor with two small hydraulic cylinders.



- Electric Motor

The primary power source is a quiet and efficient electric motor powered by a standard 230V AC (domestic) connection. This makes the scissor lift ideal for indoor environments and eliminates the requirement for dedicated generators, reducing operational complexity.

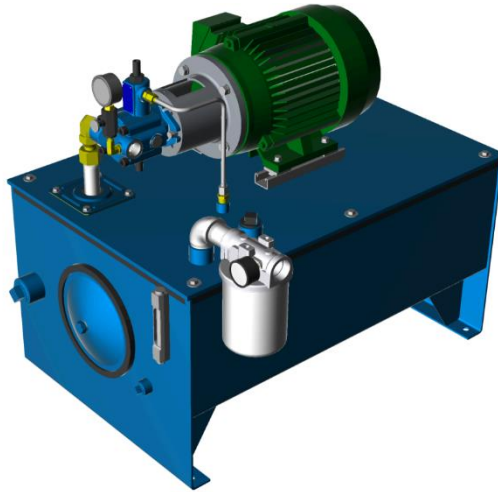


Figure 1: Hydraulic cylinders power up unit

- Hydraulic Cylinders:

Two strategically positioned hydraulic cylinders automatically engage when the platform approaches its maximum height. These cylinders provide an extra boost in lifting capacity for heavier loads, expanding the application range without incurring the cost and complexity of a full hydraulic system.

- Pressure Distribution: The hydraulic system utilizes a pressure relief valve. When activated, this valve automatically diverts pressure to the two cylinders, ensuring an even distribution of force throughout the system. This even pressure distribution reduces the overall pressure requirement, contributing to cost savings.
- Cylinder Valves: Each hydraulic cylinder features a dedicated valve for precise platform movement control:
 - * Holding Valve: This valve automatically engages to halt the platform's ascent at the desired height, preventing overextension and ensuring safe operation.
 - * Release Valve: This valve allows for controlled descent of the platform when needed, enabling smooth and efficient lowering of workers and materials.

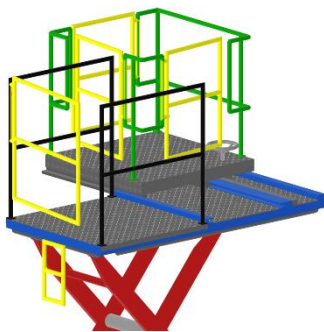


Figure 3: Fixed Safety Bench

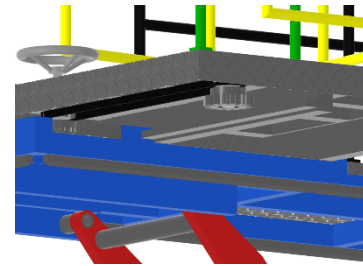


Figure 2: Horizontal Platform Movement:

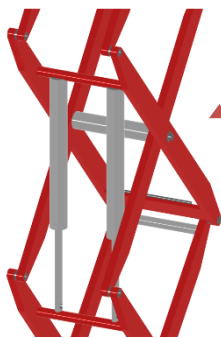


Figure 6: Hybrid Power System:

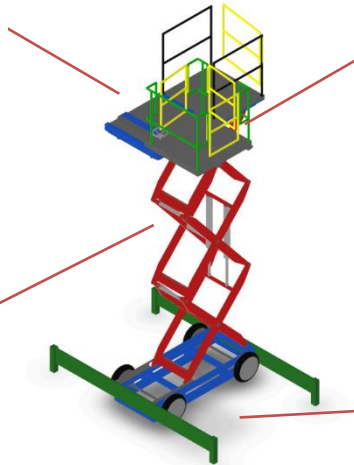


Figure 4: Main structure of my design

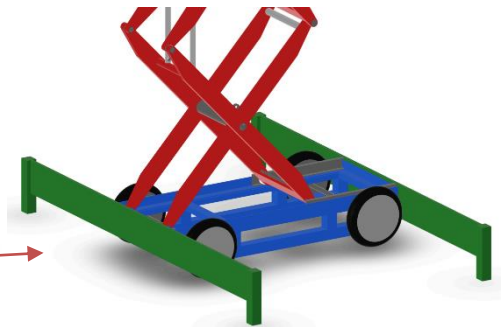


Figure 5: Improved Stability

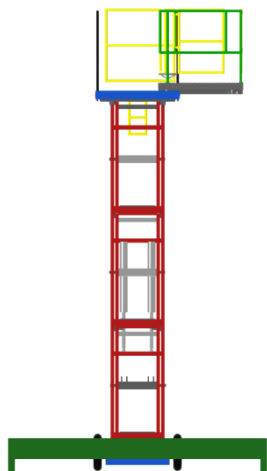


Figure 8: Front view of main structure

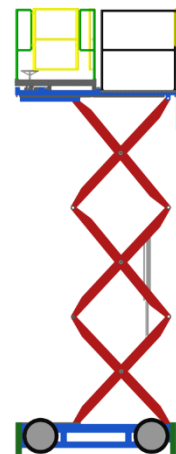


Figure 7: Right view of main structure

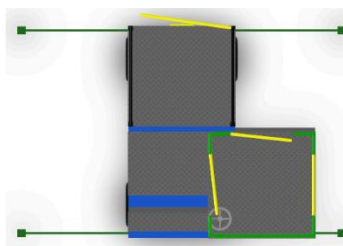


Figure 9: Top view of main structure

Future Modifications:

- **Platform Extension:** This design can be further enhanced by incorporating a telescopic platform extension. This extension would increase the work area coverage, providing greater versatility for various tasks.
- **Wireless Controls:** Replacing the hand crank with wireless controls for horizontal platform movement can be explored. This would enhance user convenience and potentially improve maneuverability within the work area.
- **Smart Safety Features:** Integrating weight sensors and overload protection systems are potential future modifications. These features would further improve safety by alerting operators of exceeding weight limitations and automatically stopping operations if necessary.

7 Designing Process and Calculations

This section details the design process and key calculations for the Enhanced Hybrid Scissor Lift, focusing on material selection and critical component sizing.

Design procedure

1. Material Selection

Chose appropriate materials for key components, considering factors like strength, weight, and cost. (Details on specific materials and justifications can be found in the "Material Selection" section of the report.)

2. Design scissor lift arms:

Determined the target lifting height and number of scissor arms for stability. Calculations were performed to define the geometry and dimensions of the scissor arms and linkages.

3. Power System:

A hydraulic power system with suitable power rating was selected from a commercial buyer. Additionally, two hydraulic cylinders were incorporated for extra lifting power at maximum height.

4. Platform and Base Design:

The platform size was calculated based on intended use and weight capacity. High-strength steel was chosen for the base for stability, and a lightweight aluminum alloy was used for the platform to reduce weight.

5. Safety and Controls:

A fixed safety bench with guardrails was incorporated for fall protection. Long outriggers were designed to extend automatically for improved stability at maximum height.

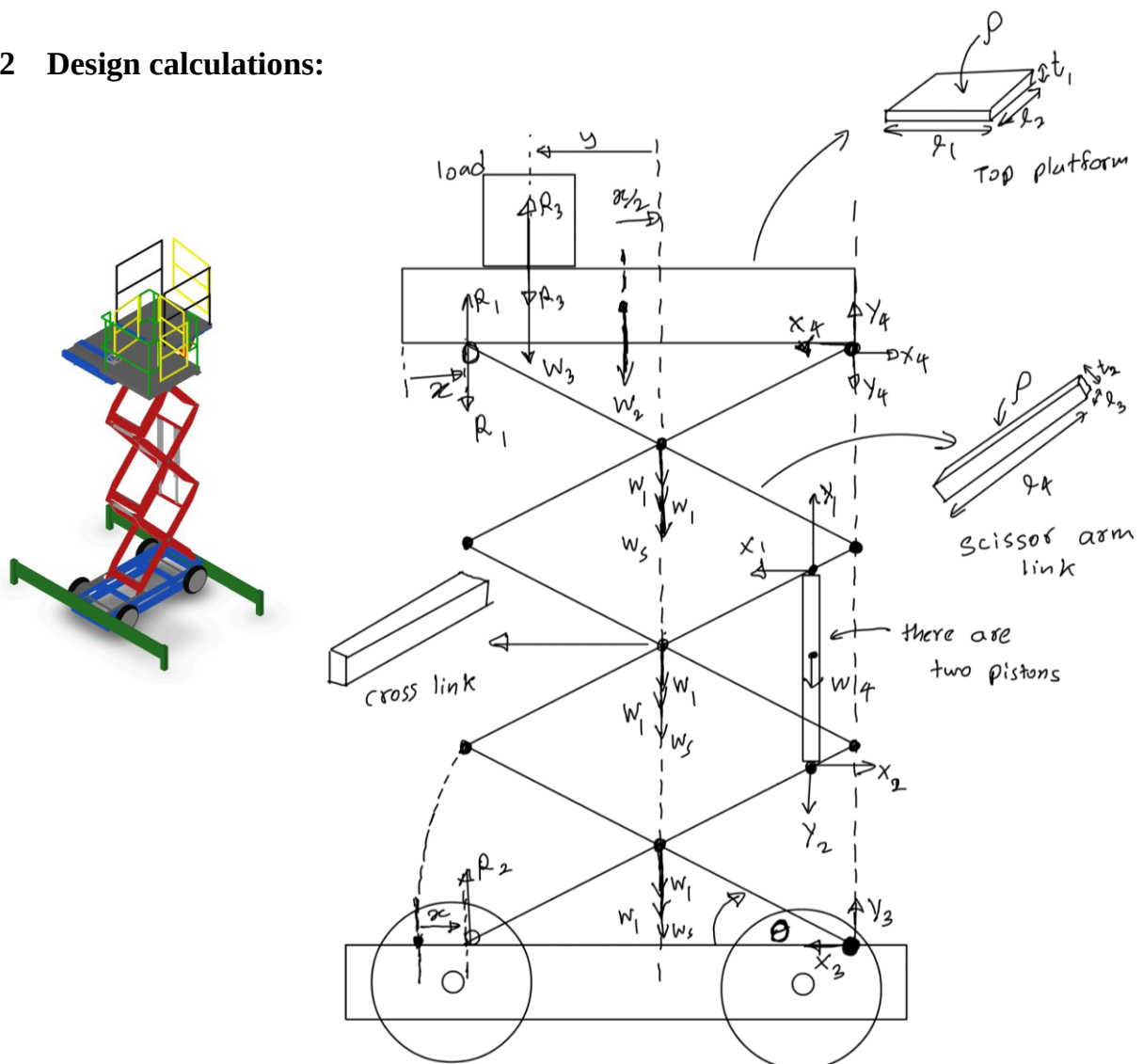
7.1 Material Selection:

The selection of materials plays a crucial role in the scissor lift's performance, weight, durability, and cost.

Component	Material	Justification
Top Platform	Aluminum Alloy (6061-T6)	Offers a good balance of lightweight construction (reducing lift weight and energy consumption) and adequate strength (typical Yield Strength = 250 MPa, Ultimate Tensile Strength = 310 MPa) for the platform.
Base Platform	High-Strength Steel (ASTM A36)	Provides a robust foundation with excellent weight-bearing capacity. High strength (Yield Strength = 255 MPa, Ultimate Tensile Strength = 480 MPa) ensures stability for the entire scissor lift.
Scissor arms	Aluminum Alloy (6061-T6)	Offers a good balance of lightweight construction (reducing lift weight and energy consumption) and adequate strength (typical Yield Strength = 250 MPa, Ultimate Tensile Strength = 310 MPa) for the platform.

Cross links	Aluminum Alloy (6061-T6)	Offers a good balance of lightweight construction (reducing lift weight and energy consumption) and adequate strength (typical Yield Strength = 250 MPa, Ultimate Tensile Strength = 310 MPa) for the platform.
Connecting Pin	High-Strength Steel (AISI 1045)	Critical for connection points due to its high strength and excellent mechanical properties. Offers suitable ultimate tensile stress (585 MPa) and shearing stress (655 MPa) for reliable operation under load.

7.2 Design calculations:



7.2.1 Scissor Lift Link Analysis

This section analyzes the forces and stresses acting on a single scissor link in the lift mechanism. The analysis assumes a simplified scenario for illustrative purposes.

Design Considerations:

- The entire load is considered to act on half the link length due to the scissor mechanism.
- The link is treated as a simply supported beam for stress calculations.
- A uniformly varying load (U.V.L.) distribution is assumed on the link.

Parameters:

- Total Link Length : $L = 1\text{m}$
- Analyzed Link Length : $L/2 = 0.5\text{m}$
- Angle Variation : $\theta = 20^\circ(\text{min}) - 70^\circ(\text{max})$
- Total Mass: mass of top platform + mass of arm links+ mass of cross links

$$\text{Total mass (W/g)} = W_2/g + 3*W_5/g + 6*W_1/g \quad g - \text{gravity}(9.81\text{ms}^{-2})$$

$$\text{Total mass} = 124.2 \text{ kg}$$

- Payload Capacity: $W_3 = 2000 \text{ N}$
- Factor of Safety: $n = 1.5$

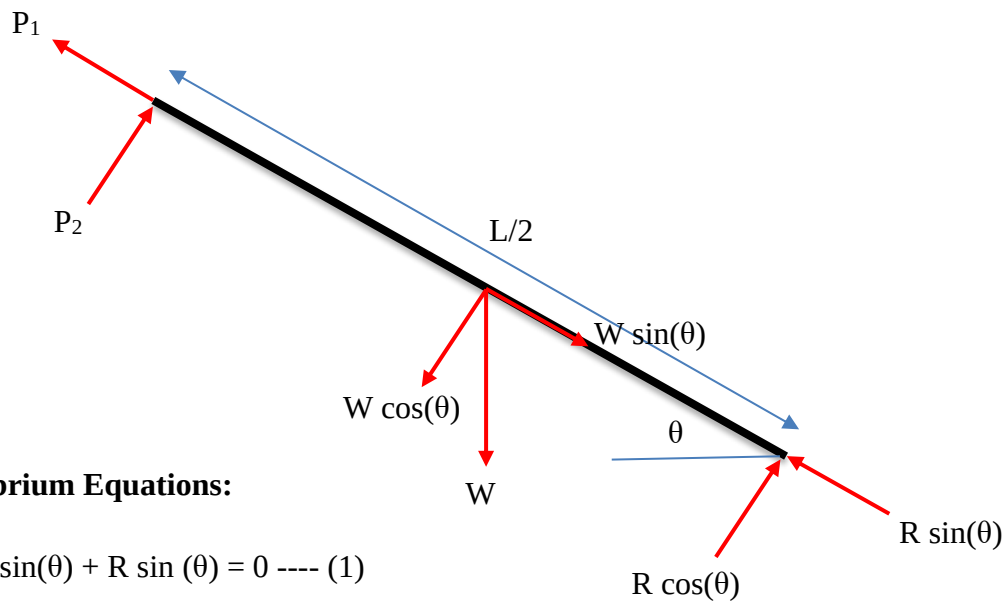
$$\text{Therefore, total mass} = 124.2 + 200*1.5 = 424.2 \text{ Kg}$$

- Total Load (P) : 4242 N
- $W = 4242/4 = 1060.5 \text{ N}$

Force Analysis:

- The total load (W) is divided into two components. ($W\cos(\theta)$ and $W\sin(\theta)$)
- Reactions at the base - Rand
- Reaction at intermediate pin is divided into two components. (P1, P2)

- When θ minimum there is a maximum load on the axes, therefore consider θ as 20° .



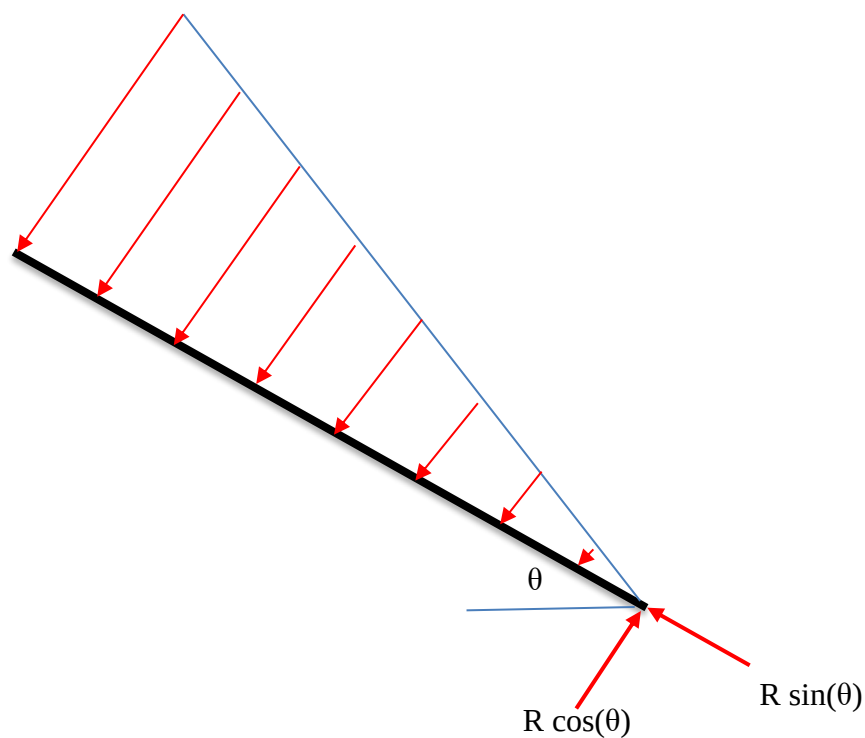
Equilibrium Equations:

$$P_1 - W \sin(\theta) + R \sin(\theta) = 0 \text{ ---- (1)}$$

$$P_2 + R \cos(\theta) - W \cos(\theta) = 0 \text{ ---- (2)}$$

Shear Force and Bending Moment:

- U.V.L. is converted to an equivalent concentrated load for shear force and bending moment calculations.
- The maximum bending moment (M_b) occurs at the point of zero shear force.



$$W \cos(\theta) = (\frac{1}{2}) * L / 2 * UVL$$

$$UVL = 3986.18 \text{ Nm}^{-1}$$

Taking moment about point A

$$P_2 * L / 2 - [2/3 * L / 2 * W \cos(\theta)] = 0$$

$$P_2 = 664.36 \text{ N}$$

By (2)

$$R = (W \cos(\theta) - P_2) / \cos(\theta)$$

$$R = 353.5 \text{ N}$$

By (1)

$$P_1 = (W - R) \sin(\theta)$$

$$P_1 = 241.8 \text{ N}$$

Stress Calculations:

1. Material Selection:

I choose Aluminum Alloy (6061-T6) as materials to top platform, scissor arm links and cross link.

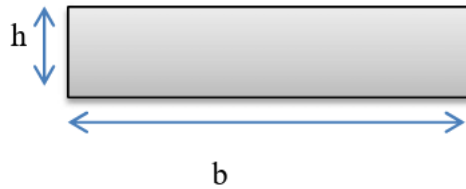
$$\sigma_{\text{allow}} = \text{Ultimate tensile strength} / \text{Factor of Safety}$$

$$\text{Ultimate tensile strength} = 310 \text{ MPa}$$

$$\sigma_{\text{allow}} = 206.67 \text{ MPa}$$

2. Moment of Inertia (I):

The moment of inertia (I) depends on the link's cross-sectional geometry. Since assuming a rectangular cross-section, the formula for the moment of inertia about the horizontal axis is:



$I = (b * h^3) / 12$, b - Width of the link (meters) , h - Thickness of the link (meters)

Assume, $Y = h/2$ and $b = 2h$

$$I = bh^3/12 = h^4/3$$

3. Maximum Bending Moment (M_b):

The maximum bending moment (M_b) occurs at the point of zero shear force, which is typically located at the midpoint of the analyzed link length ($L/2$). Therefore:

$$M_b = (W \cos(\theta) * (L/2)^2) / 8 = 31.14 \text{ Nm}$$

4. Bending Stress (σ):

Utilize the formula for bending stress:

$$\sigma_{\text{allow}} = M_b * Y / I$$

Y : Distance from the neutral axis to the outermost fiber (m) - This is typically half the thickness ($h/2$) for a rectangular cross-section.

$$206.67 * 10^6 = 31.14 * (h/2) / (h^4/3)$$

$$h = 6.1 \text{ mm} , b = 12.2 \text{ mm}$$

7.2.1 Design of moving end pins

Allowable Shear Stress:

- Material (Aluminum Alloy 6061-T6):

Ultimate Tensile Strength (σ_{ut}) = 310 MPa

- Factor of Safety (n) = 2 (common for shear stress)
- Allowable Shear Stress (τ_{allow}) = $\sigma_{ut} / (2 * n) = 310 \text{ MPa} / (2 * 2) = 77.5 \text{ MPa}$

Minimum Required Pin Diameter (D):

- Torque (T):

Calculated torque value of $T = 200 \text{ Nm}$

Polar Moment of Inertia (J) for a solid circular cross-section: $J = (\pi * D^4) / 32$

Using the Torsion Formula:

- $\tau_{allow} = T / J$
- Rearranging to solve for D:

$$D \geq [(32 * T) / (\pi * \tau_{allow})]^{(1/4)}$$

- Substitute the values:

$$D \geq [(32 * 200 \text{ Nm}) / (\pi * 77.5 \text{ MPa} * 10^6)]^{(1/4)} \approx 7.2 \text{ mm}$$

Standard Pin Selection:

As mentioned, $D = 10 \text{ mm}$ is a common choice, and it provides a margin of safety compared to the calculated minimum ($D \geq 7.2 \text{ mm}$).

8. Cost Analysis

1. Material Costs:

- Aluminum Alloy (6061-T6):
 - Cost per kg: \$2.50 USD (Approximately 500 LKR)
- High-Strength Steel (ASTM A36):
 - Cost per kg: \$1.00 USD (Approximately 200 LKR)

- High-Strength Steel (AISI 1045):
 - Cost per kg: \$1.20 USD (Approximately 240 LKR)
- 2. Manufacturing Processes:
 - Machining (cutting, drilling, milling): \$50 USD per hour (Approximately 10,000 LKR per hour)
 - Welding (including labor and consumables): \$60 USD per hour (Approximately 12,000 LKR per hour)
- 3. Component Costs:
 - Electric Motor: \$500 USD (Approximately 100,000 LKR)
 - Hydraulic Cylinders (x2): \$300 USD each (Approximately 60,000 LKR each)
 - Fixed Safety Bench with Guardrails: \$200 USD (Approximately 40,000 LKR)
 - Long Outriggers: \$150 USD (Approximately 30,000 LKR)
 - Connecting Pins (High-Strength Steel): \$5 USD each (Approximately 1,000 LKR each)
- 4. Additional Features:
 - Horizontal Platform Movement Mechanism: \$100 USD (Approximately 20,000 LKR)
 - Wireless Controls (Optional): \$150 USD (Approximately 30,000 LKR)
 - Smart Safety Features (Weight Sensors, Overload Protection): \$200 USD (Approximately 40,000 LKR)

Now, let's calculate the total cost of materials and components:

1. Material Costs:
 - Aluminum Alloy (6061-T6) for Top Platform, Scissor Arms, and Cross Links:
 - Total weight estimation: Approximately 50 kg

- Cost: $50 \text{ kg} * 500 \text{ LKR/kg} = 25,000 \text{ LKR}$
- High-Strength Steel (ASTM A36) for Base Platform:
 - Total weight estimation: Approximately 100 kg
 - Cost: $100 \text{ kg} * 200 \text{ LKR/kg} = 20,000 \text{ LKR}$
- High-Strength Steel (AISI 1045) for Connecting Pins:
 - Total number of pins: Approximately 10
 - Cost: $10 \text{ pins} * 1,000 \text{ LKR/pin} = 10,000 \text{ LKR}$

Total Material Costs = $25,000 \text{ LKR} + 20,000 \text{ LKR} + 10,000 \text{ LKR} = 55,000 \text{ LKR}$

2. Component Costs:

- Electric Motor: 100,000 LKR
- Hydraulic Cylinders (x2): $2 * 60,000 \text{ LKR} = 120,000 \text{ LKR}$
- Fixed Safety Bench with Guardrails: 40,000 LKR
- Long Outriggers: 30,000 LKR
- Connecting Pins (High-Strength Steel): $10 \text{ pins} * 1,000 \text{ LKR/pin} = 10,000 \text{ LKR}$

Total Component Costs = $100,000 \text{ LKR} + 120,000 \text{ LKR} + 40,000 \text{ LKR} + 30,000 \text{ LKR} + 10,000 \text{ LKR} = 300,000 \text{ LKR}$

3. Additional Features:

- Horizontal Platform Movement Mechanism: 20,000 LKR
- Wireless Controls (Optional): 30,000 LKR
- Smart Safety Features (Weight Sensors, Overload Protection): 40,000 LKR

Total Additional Features Costs = $20,000 \text{ LKR} + 30,000 \text{ LKR} + 40,000 \text{ LKR} = 90,000 \text{ LKR}$

Now, let's calculate the total cost of manufacturing processes:

- Assuming 50 hours of machining and 20 hours of welding: Total Machining Cost = 50 hours * 10,000 LKR/hour = 500,000 LKR Total Welding Cost = 20 hours * 12,000 LKR/hour = 240,000 LKR

Total Manufacturing Costs = 500,000 LKR + 240,000 LKR = 740,000 LKR

Finally, let's calculate the overall total cost:

Total Cost = Total Material Costs + Total Component Costs + Total Additional Features Costs + Total Manufacturing Costs = 55,000 LKR + 300,000 LKR + 90,000 LKR + 740,000 LKR = 1,185,000 LKR

Therefore, the estimated total cost of designing and manufacturing the enhanced hybrid lifting system scissor lift, including materials, components, additional features, and manufacturing processes, is approximately 1,185,000 LKR.

9. SWOT analysis

Strengths:

1. **Versatility:** The hybrid design combines the quiet operation of an electric motor with the lifting power of hydraulic cylinders, catering to a wide range of industrial needs without the high cost of a full hydraulic system.
2. **Enhanced Stability:** Longer beams and outriggers improve stability, especially at maximum platform heights, ensuring safer working conditions for operators.
3. **Safety Features:** The incorporation of a fixed safety bench with guardrails provides continuous fall protection, prioritizing worker safety throughout operation.
4. **Ease of Use:** The design includes intuitive controls for horizontal platform movement, reducing the need for repositioning the entire scissor lift for minor adjustments, thereby increasing efficiency and user convenience.
5. **Environmental Considerations:** The electric motor reduces noise pollution and emissions, making it suitable for indoor environments and contributing to a cleaner working environment.

Weaknesses:

1. **Initial Cost:** The hybrid system and additional safety features may lead to a slightly higher initial cost compared to traditional scissor lift designs.
2. **Complexity:** The incorporation of multiple systems (electric motor and hydraulic cylinders) adds complexity to the design, potentially increasing maintenance requirements and troubleshooting efforts.

Opportunities:

1. **Market Expansion:** The versatility and safety features of the design may attract new customers or industries seeking more efficient and safer lifting solutions.
2. **Technological Advancements:** Continued research and development in electric motors and hydraulic systems could lead to further enhancements in efficiency, performance, and cost-effectiveness of the hybrid design.

Threats:

1. **Competitive Landscape:** Competitors may introduce similar or more advanced designs, posing a threat to market share and profitability.
2. **Regulatory Changes:** Changes in safety regulations or environmental standards could impact the design requirements and market viability of the hybrid scissor lift.

10. References

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Annexes

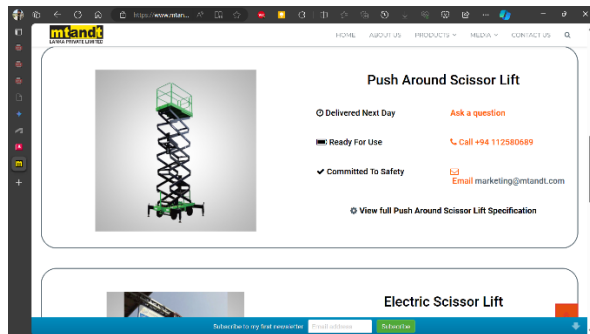


Figure 1: <https://www.mtandt.lk/scissor-lift-manufacturers/>

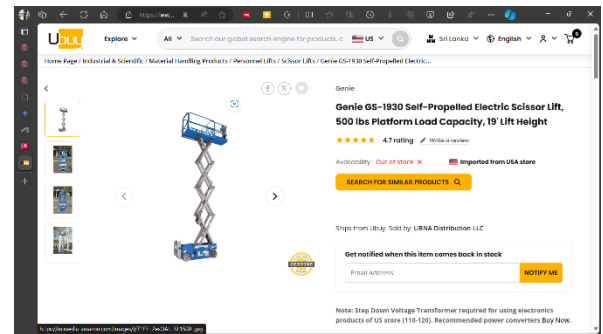


Figure 2: <https://www.ubuy.com.lk/en/product/42RE88ITS-5-star-16-feet-max-lift-new-electric-scissor-lift-man-lift-skylift-jack>

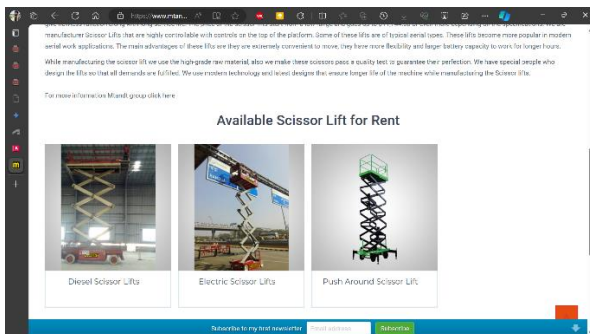


Figure 3: <https://www.mtandt.lk/scissor-lift-manufacturers/>

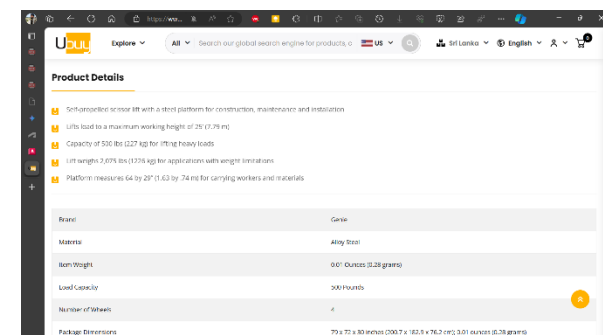


Figure 4: <https://www.ubuy.com.lk/en/product/42RE88ITS-5-star-16-feet-max-lift-new-electric-scissor-lift-man-lift-skylift-jack>

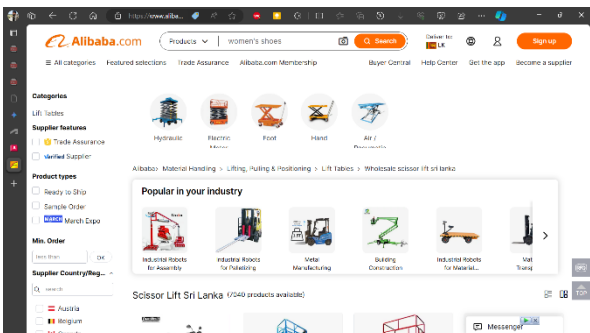


Figure 5: <https://www.alibaba.com/showroom/scissor-lift-sri-lanka.html>

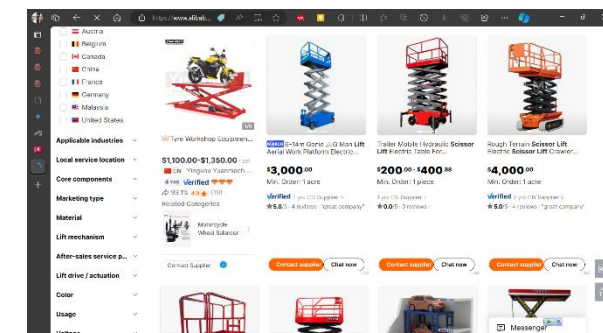


Figure 6: <https://www.alibaba.com/showroom/scissor-lift-sri-lanka.html>